

HAT-MedMNIST

A hybrid agentic training framework for MedMNIST

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Agentic AI / medical-image benchmarks

- 1 **Motivation, Objectives & Plan**

- 2 **System Architecture & MVP**

- 3 **Prior Art, Governance & Validation**

What it is

A **modular, hybrid agentic framework** for training neural networks on MedMNIST.

- Role-specialised agents own **one duty each**.
- They interact **only through versioned artefacts**.
- An independent reviewer agent **audits every stage and can veto**.

The contribution

The orchestration is the contribution — **not** the classifier.

Conventional pipelines

- Tightly coupled.
- Hard to reproduce and audit.

Typical LLM-agent frameworks

- Non-determinism everywhere.
- Locked to one proprietary model.

HAT-MedMNIST

- Roles *plus* explicit contracts.
- Non-determinism confined to declared decisions.
- Every choice is traceable.
- An independent automated reviewer.

- Cohesive, weakly-coupled agents that talk only via typed artefacts.
- A **representation-architect** agent that reasons about the best input.
- A **hybrid reasoning seam**: open models with heuristic fallback.
- **Ambiguity handling** and leakage-safe data splits.
- **Abstention** on the “no confident finding” case.
- An integrated MVP demonstrated at **TRL 7**.

WP	Focus
WP1	Requirements & architecture
WP2	Data understanding, representation & prior-art scouting
WP3	Hybrid orchestration & prior-art-informed model design
WP4	Evaluation, enrichment & comparison
WP5	MVP integration & validation
WP6	Consistency & anomaly governance (cross-cutting)

Indicative 12-month plan

WP / Month	1	2	3	4	5	6	7	8	9	10	11	12
WP1	•	•										
WP2		•	•	•								
WP3			•	•	•	•	•					
WP4					•	•	•	•	•			
WP5								•	•	•	•	•
WP6		•	•	•	•	•	•	•	•	•	•	•

WP6 is cross-cutting across the whole timeline.

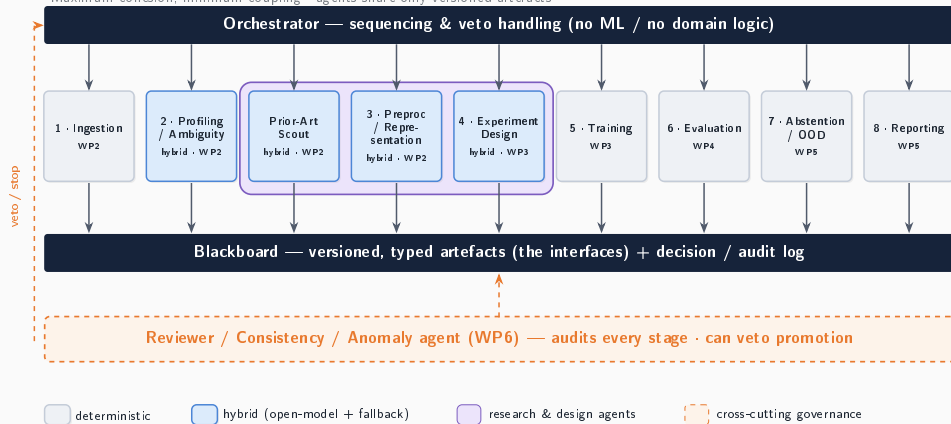
Risk plan

Risk	Impact / Prob.	Mitigation
Data leakage / invalid split	High / Med	Train-only stats, seeded splits, reviewer gate before training
Non-deterministic LLM reasoning	Med / Med	Temperature 0, heuristic fallback, logged provenance
Agent contract failure	High / Med	Typed artefact contracts, orchestrator veto, fault-injection tests
Performance below baseline	High / Med	Baseline-first strategy, ablations, documented negative results
Vendor lock-in	Med / Low	Open-model seam, no proprietary dependency

The Blackboard architecture

Hybrid Agentic MedMNIST — architecture

Maximum cohesion, minimum coupling · agents share only versioned artefacts



- **No agent calls another.**
- They share only versioned artefacts on a **Blackboard**.
- A stateless **Orchestrator** sequences them.

The claim, made testable

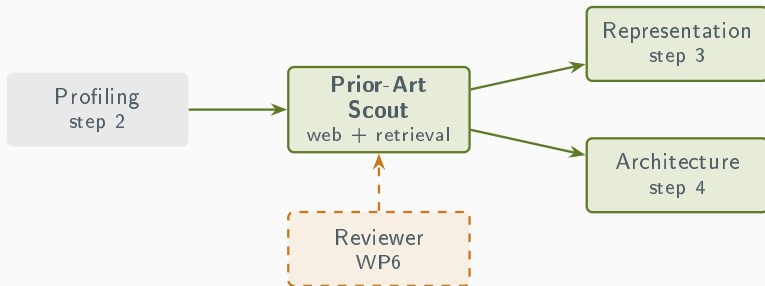
Any agent can be replaced or upgraded without touching another — **zero direct inter-agent calls.**

The agents

Agent	Duty	Kind
Orchestrator	sequencing, veto handling	deterministic
Ingestion	load MedMNIST	deterministic
Profiling / Ambiguity	profile data, flag ambiguity	hybrid
Prior-Art Scout	search literature, propose ideas	hybrid
Preprocessing	architect the representation	hybrid
Experiment Design	pick the configuration	hybrid
Training	train the selected model	deterministic
Evaluation	test metrics	deterministic
Abstention / OOD	refuse when unconfident	deterministic
Reporting	assemble the dossier	deterministic
Reviewer (WP6)	consistency / anomaly, veto	hybrid

The Prior-Art Scout

A hybrid component of the pipeline: it reads the literature and feeds **cited ideas** into representation (step 3) and architecture (step 4).



It *advises* the hybrid decisions — it never trains, selects, or calls another agent.

The hybrid approach

Deterministic execution

- data, splits, training, metrics, ranking.

Open-model reasoning, only at:

- ambiguity assessment,
- representation design,
- configuration choice,
- consistency review.

No vendor lock-in

Plug in a local open model. With none, it falls back to heuristics and still runs offline and reproducibly.

Reproducible reasoning

Every decision records its source; identical seeds and configs replay the run.

- Runs end-to-end on **PathMNIST** (nine-class histology).
- Preserves the **official split**; the test set is a different clinical centre.
- Agents **search and train** bounded architectures automatically.
- Compared against a **conventional baseline** on the same data and seeds.

Leakage-safe by design

- The search loop **never sees test metrics**.
- Selection is driven by **validation only**.
- The winner is frozen, then evaluated on test **once**.
- Repeated seeds quantify run-to-run variance.

Independent audit

A data-quality and leakage audit must pass **before** any training begins.

From literature to decisions

Two levels, at very different cost:

- **Advise within bounds** — rank & justify existing options with evidence. Cheap, safe, immediate.
- **Propose new options** — outside the current set; promoted through a gate before anything runs.

Representation (step 3)

New label-preserving augmentations: validated, then whitelisted — **semi-automatic**.

Architecture (step 4)

A new family stays a **reviewed spec** — implemented as a registry entry, then runnable.

Reproducible & governed

Reproducible by design

- Snapshot + checksum every source.
- A rerun replays the cache.
- Temperature 0; heuristic fallback when offline.

Governed (WP6)

- The Reviewer verifies every citation resolves.
- Claims must be backed by a source.
- It can **veto** the brief.

Measured

100% of ideas cited · 0 hallucinated citations · brief reproducible from the cache.

- An **independent reviewer** audits each stage.
- It can request a **retry** or **veto** promotion.
- It checks evidence, traceability and reproducibility.

Measured

≥90% of injected inconsistencies detected · 100% of decisions logged · full run reproducible.

System prototype in a representative environment

An end-to-end workflow — ingestion, representation, coordinated decisions, training, evaluation, abstention, auditing — demonstrated by someone other than the implementer.

- Research demonstrator on **benchmark data**.
- **No clinical claims**; uncertain cases go to human review.

Final acceptance criteria

- All mandatory deliverables complete, versioned, and linked to requirements.
- KPIs measured with documented procedures and objective evidence.
- End-to-end hybrid workflow demonstrated, including abstention on uncertain cases.
- Results reproducible from stored configs, code, data references and seeds.
- No critical risk remains without an approved mitigation.

Reviewer sign-off

Confirms coherence of the scientific advancement, industrial potential, and TRL 7 evidence.

Thank you

Maximum cohesion, minimum coupling.